Bihar Engineering University, Patna
End semester Examination – 2023
Semester-VI

Course: B.Tech. | *Time: 03 Hours*
Code: 100606 | *Subject: Digital Signal Processing* | *Full Marks: 70*

Instructions:-
(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 *Answer any seven question only:* **[2 x 7 = 14]**

(a) Define discrete-time system.
(b) Write the significance of ROC.
(c) What is twiddle factor?
(d) Write the difference between Causal and anti-causal system.
(e) Define Cut-off frequency.
(f) What is sampling theorem?
(g) What do you mean by transposed form structure?
(h) Write the properties of unit impulse response.
(i) Write the difference between one-sided and two-sided Z-transform.
(j) What is Signal Processing?

Q.2 (a) Derive the expressions for order and cut-off frequency of a Butterworth filter. [7]
(b) State and prove time reversal property of DFT. [7]

Q.3 (a) Explain frequency domain representation of sampling. [7]
(b) Explain how bandlimited signal can be reconstructed from its sample? [7]

Q.4 Compare computational complexity using direct DFT computation and using FFT. Compute the 8-point DFT of the sequence [14]
$X(n)=1\ ,\ 0 \le n \le 7$
$=0\ ,$ otherwise
Use DIT-FFT algorithm. (Show flow graph)

Q.5 (a) Write the steps in designing Chebyshev filter? [7]
(b) How can you design a digital filter from analog filter? [7]

Q.6 (a) What are the advantages and disadvantages of FIR filter? [7]
(b) For what kind of application, the symmetrical and anti symmetrical impulse response can be used? [7]

Q.7 (a) Determine the region of convergence of right-sided,left-sided and finite duration two-folded sequences. [7]

(b) Using long division, determine the inverse z-transform of $x(z)=\frac{1+2z^{-1}}{1-2z^{-1}+z^{-2}}$,if [7]
(i) x(n) is causal.
(ii) x(n) is anti-causal.

Q.8 (a) What are the applications of Digital Signal processing? Explain in details. [7]

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(b) Find the Nyquist rate and Nyquist interval for the continuous-time signal given below: [7]

$$X(t)=\frac{1}{2\pi}\cos(4000\pi t)\cos(1000\pi t)$$

Q.4 Write short notes on *any two* of the following:- [7 x 2 =14]

(a) Gibbs Phenomenon.

(b) Wiener filter.

(c) Park-McClellan's method.

(d) Low pass and Band pass filter.

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